

LESSON 9.2

STATISTICAL QUESTIONS



LESSON INTRODUCTION

MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.

LEARNING TARGETS

- I can write statistical questions that generate numerical data.

BENCHMARK COHERENCE

BUILDING ON

MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- What is a statistical question?
- What is needed in a survey question to generate numerical data?

LESSON NARRATIVE

In this lesson, students will create statistical survey questions to gather data from the class. Students will classify the data as categorical or quantitative. Students may encounter non-statistical questions and have the chance to identify them and rewrite them as a statistical question.

COMPONENT SUMMARY

9.2.1 WARM-UP (~5 MIN)

Create survey questions

9.2.2 ACTIVITY (25-30 MIN)

Gather data through survey questions

9.2.3 YOUR TURN! (10 MIN)

Examples and non-examples of statistical questions

9.2.4 WRAP-UP (~5 MIN)

Reflect on ability to reach learning target

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Statistical question

MATERIALS

- (Optional) Chart paper
- (Optional) Chart makers
- (Optional) Computers

ADDITIONAL PREPARATION

- If using Forms or Excel, consider teaching students how to use them before this lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may still need help identifying statistical and non-statistical questions or rewriting them.
- Responses may be imprecise, or questions may be misread.

THINGS TO CONSIDER

- It should become evident that the accuracy or precision of the data is important for it to be used effectively. Answering that “I have more than 1 sibling” is not as useful as saying “I have 3 siblings.”

LITERACY AND LANGUAGE ROUTINES

Co-Craft Questions and Problems

This routine encourages students to produce and craft mathematical questions themselves. The 9.2.1 Warm-Up and 9.2.2 Activity require students to work in pairs or small groups to create statistical questions, allowing students the opportunity to actively participate in the survey process to deepen understanding and connect mathematics to their own lives.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In the 9.2.1 Warm-Up, students will create real-life questions to ask their classmates and gather data. In creating their own questions, students will feel more connected to the data and have a chance to connect the learning to their everyday lives.

DIFFERENTIATE INSTRUCTION

9.2.2 Activity provides multiple means for students to engage with the content of the lesson. Students discuss statistical questions in partners or small groups to determine the best question(s) to pose to their classmates, providing auditory entry points and engagement through verbal discussion. The gallery walk encourages students to actively participate in the learning as they move throughout the classroom to discuss and answer statistical questions, encouraging mathematical discourse and investment in the content.

9.2.1 WARM-UP: MATH TALK

Component Overview: In this Warm-Up, students will create questions to later survey the class with to gather statistical data. Consider reviewing student answers and having students revise their questions if they do not meet the criteria.

TEACHER MOVES

Students may struggle to create statistical questions. Check student answers for appropriateness. Consider allowing students to brainstorm ideas for numerical data they can find out about their classmates before forming their questions.

DIFFERENTIATE INSTRUCTION

MA.K12.MTR.7.1: Real-World Contexts

In this Warm-Up, students will create real-life questions to ask their classmates and gather data. In creating their own questions, students will feel more connected to the data.



9.2.1 Warm-Up: Math Talk

In the spaces below, write three statistical questions that you could use today to collect data from your classmates. Be sure at least one of your questions will collect numerical data. *Answers will vary.*

1. How old are the students in my class?

2. How tall, in inches, are the students in my class?

3. How long does it take for my classmates to get to school in the morning ?



9.2.2 Activity: Gallery Walk

1. With your partner, discuss the following questions.
- a. How do you know the questions you wrote are statistical questions?
Questions are statistical if you can use them to collect data with variability.
 - b. How can you determine which of the questions will generate numerical data?
The questions will yield a variety of numerical responses.
2. With your partner, determine the two best questions whose answers provide **numerical data** from the six questions you each created.

Determine which of these two questions each partner will use to collect data about your class.

Answers will vary.

3. Use your statistical question to collect some numerical data following the directions below. This data will be used in a later lesson, so keep it safe.
- a. Write your statistical question in the top of the data chart that you chose in the previous problem.
 - b. Place the data chart in a location that is easily accessible by your classmates.
 - c. Follow your teacher's directions to answer each group's question.
Answers will vary.

My Statistical Question			
How tall in inches are the students in my class?			
Name	Answer	Name	Answer
1. Reid	64 inches	15.	
2. Kaylee	63 inches	16.	
3. Ava	60 inches	17.	
4. Sebastian	58 inches	18.	
5.		19.	
6.		20.	
7.		21.	
8.		22.	
9.		23.	
10.		24.	
11.		25.	
12.		26.	
13.		27.	
14.		28.	

9.2.2 ACTIVITY: GALLERY WALK

Component Overview: In this Gallery Walk, students work with a partner to choose the best statistical question from the 9.2.1 Warm-Up to pose to the class. Each partner will choose a question and then post it in a location around the room for the rest of the class to answer. Ensure students understand where and how to post their questions. The class will rotate to answer each question. If you have a large class, consider placing students in groups and posting one question per group.

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems for students to frame their thinking when discussing questions with their partner:

- “Questions are statistical if they...”
- “I know questions are statistical because they...”
- “Questions will generate numerical data if the responses are...”

TEACHER MOVES

Ensure students are clear on where to post their questions throughout the room and where and when to move to each question. Consider how you will signal the class to move on, how many students should be at each chart, and how each group should be moving around the room.

9.2.3 YOUR TURN!

Component Overview: In this Your Turn!, students give examples of statistical and non-statistical questions. Students explain the definition of a statistical question in their own words, then use the definition to justify whether sample questions are statistical or not.

TEACHER MOVES

When discussing non-examples, encourage students to think of questions that could not have multiple answers. Students should also consider questions that are numerical but only have one response. Connect the word “statistical” with stats, like in sports.

ENRICHMENT

Students may struggle to explain their responses in writing. To further scaffold, ask questions like:

- What makes a question a statistical question? Why?
- How many answers does the question about the man’s height have? Will it have more than one?
- Does everyone live the same distance from school? What does that tell you about the question?



9.2.3 Your Turn!

1. Give two examples and two counterexamples of a statistical question.

Examples of Statistical Questions	Counterexamples that are NOT Statistical Questions
<i>How often do you go to the public library?</i>	<i>Which state has the most public libraries?</i>
<i>What is the average age of U.S. presidents at their inauguration?</i>	<i>What is the age of the youngest U.S. president at the time of inauguration?</i>

2. Write the definition of a statistical question in your own words.
Answers will vary. Statistical questions can be answered by collecting data that varies.
3. The question, “How tall is the tallest man in the world, in inches?” is not statistical. Use your definition of a statistical question to justify why it is not a statistical question.
There is only one correct answer to the question. It does not elicit a variety of responses.
4. Decide if the question, “How many miles do you live from the school campus?” is statistical or non-statistical. Explain your decision.
It is statistical because you can use it to gather data with a variety of responses.



9.2.4 Wrap-Up: Reflection

1. What is the most difficult part about writing statistical questions?
Answers will vary. Figuring out if the question elicits a numerical response or not.
2. How can it be determined whether or not a statistical question can be answered with numerical data?
*Answers may vary.
A statistical question can be answered with numerical data if data has numbers that are not used as categories.*
3. What was the most surprising response to your statistical question? Why?
Answers will vary. Responses should apply to the statistical question the student created.

9.2.4 WRAP-UP: REFLECTION

Component Overview: In this Wrap-Up, students will discuss how they felt throughout the lesson trying to reach the learning target. Consider using this as a guide for the next day to see what to focus on or reteach with each student.

DIFFERENTIATE INSTRUCTION

Some students may struggle to explain their reflections in writing. Consider alternate means of assessment. Students could partner up and answer the questions verbally as you circulate and listen for common themes.